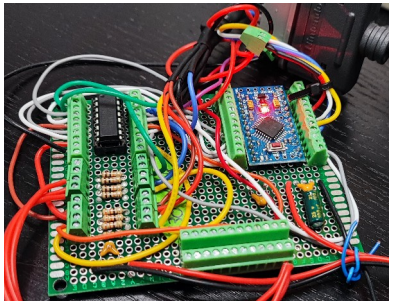
SNEngine ver. 20260414-1
USER MANUAL ver. 2

ABOUT SNENGINE

SNEngine is meant to be a high-performance, near-zero lag hardware adapter designed to connect original or third-party SNES controllers (e.g. SN30 2.4GHz) to the PC Engine console family. While designed for original wired SNES controllers, this adapter is fully compatible with the 8BitDo Retro Receiver for SNES. This allows you to use virtually any modern Bluetooth gamepad on your PC Engine (e.g. DS4 or DS5). Please note: The cases you can see on the pictures below are taken from the SNES-to-NeoGeo project. These are ideal to house SNEngine's electrical components.

Optional OLED Display



SNEngine Prototype Board

SNEngine using the SNES-to-Neo-Geo projects' case

NEW IN THIS VERSION

- **Controller button display option for OLED:** If you want to connect an XBOX or PSX controller via RetroReceiver or another BT dongle, you can use the hotkey combo SELECT+LEFT to display the correct button names for your controller. This has resulted in a slightly different arrangement of text on the display.

The 2nd row is now showing the controller type and the mapping moved to the 3rd.

Notifications now only occupy one row instead of two.

Key Features

- **Dynamic Button Mapping:** Choose from 8 distinct controller layouts to match your playstyle or specific game requirements.
- **Multi-Stage Autofire (Turbo):** Integrated turbo functionality for PCE buttons I and II with 6 adjustable speed tiers, ranging from 11 Hz to 31 Hz. The mapping logic allows you to hold a dedicated standard button to charge shots instantly, even while Auto-Fire is active on the same PCE input.
- **On-the-Fly Configuration:** Switch between layouts and turbo speeds instantly using simple Hotkey combos (SELECT + D-Pad).
- **Persistent Memory:** All settings (current layout and turbo speed) can be saved to the EEPROM, ensuring your preferences are ready as soon as you power on the console.
- **Optional OLED display support**

NEXT DEVELOPMENT STAGES

I am also exploring **6-button support**. Since this requires some core hardware and software changes, it will take some more time to finalize. Thanks for following the project!

BUTTON LAYOUTS

You can choose your preferred button layout by pressing **SELECT + DOWN** simultaneously.

If you've implemented the optional OLED display, you can press **SELECT + LEFT** to switch the button labels (SNES, Xbox, and PSX). This can be useful if you want to connect, for example, a DS4 or an 8BitDo Arcade Stick (using Xbox-style button labels) via the RetroReceiver. Please note that, due to limitations in the character set, the PSX button symbols are represented as alphabetic characters:

Cross=X, Circle=O, Square=S, Triangle=T

BUTTON LAYOUTS

Controller Type	SNES	A	B	X	Y	L	R
	XBOX	B	A	Y	X	L	R
	PSX	O	X	△	□	L	R
Button Layout	A	2	1	1	2	T-1	T-2
	B	1	2	2	1	T-2	T-1
	C	2	1	T-2	T-1	1	2
	D	1	2	T-1	T-2	2	1
	E	T-2	T-1	1	2	1	2
	F	T-1	T-2	2	1	2	1
	G	2	T-2	1	T-1	1	2
	H	1	T-1	2	T-2	2	1

* T=Turbo

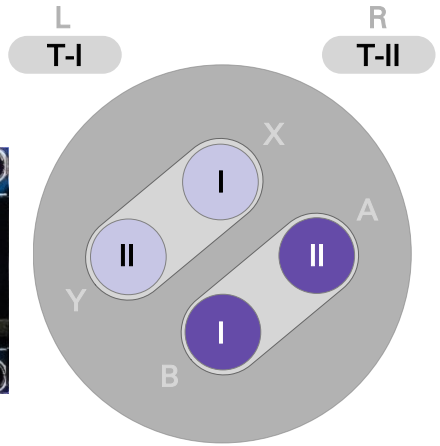
** **The 8BitDo M30 (2.4G version)** can be connected using the 2.4G adapter from the SN30. The button mapping corresponds to that of an SNES controller (A=A, B=B, …) , with the exception that Z functions as the left shoulder (L) button, C replaces the right shoulder (R) button and Minus (-) acts like SELECT. Of course, you can connect the M30 bluetooth version to the retro receiver and have the usual shoulder button mapping.

LAYOUT EXAMPLES

Assigning autofire to a shoulder button like in Layout A makes particular sense if you want to perform a charge shot with a different button. If you need autofire on both buttons anyway, you're better off using a different mapping, such as Layout H.

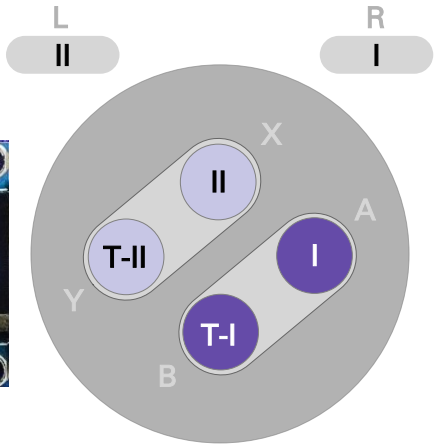
LAYOUT A

Recommended for: World Sports Competition/Power Sports, R-Type



LAYOUT H

Recommended for: Salamander



AUTOFIRE SPEED

You can adjust the autofire frequency by pressing the hotkey combination **SELECT+UP**.

- Setting 1: ~11.36 Hz (works great for Bloody Wolf)
- Setting 2: 12.50 Hz (nice for R-Type & Salamander)
- Setting 3: ~13.89 Hz
- Setting 4: ~15.63 Hz
- Setting 5: ~20.83 Hz
- Setting 6: 31.25 Hz (optimal for Makyou Densetsu / The Legendary Axe)

AUTOFIRE OVERRIDE

Signals from autofire buttons are overridden by pressing the "normal" buttons with the same number. For example, if you hold down autofire 1 and then press button 1, the signal from button 1 will take effect. The same applies to autofire 2 in conjunction with button 2. This is useful for games where you want to use both - autofire and charge shots. This kind of gameplay is really awkward to achieve with the official PC Engine controllers, since you have to activate and deactivate a special autofire toggle switch. Games where this is advantageous include:

- Aldynes (SuperGrafx),
- 1941 Counter Attack (SuperGrafx),
- Cotton – Fantastic Night Dreams,
- Detana! TwinBee,
- Gradius I & II (long laser beam, not real charge),
- Jinmu Denshou,
- Rayxanber II,
- R-Type I & II / R-Type Complete CD,
- Salamander (long laser beam, not real charge),
- Sinistron (AKA Violent Soldier).

SAVING THE SETTINGS

You can store your chosen button layout and autofire setting by pressing **SELECT+RIGHT**.

The settings will not be rewritten to the EEPROM unless they have changed since the last save.

- **Code & diagrams:** https://github.com/retrocityransom/snengine
- **Trailer:** https://youtu.be/9fGX9Tb3N7Q
- **Robin Edwards' SNES-to-NeoGeo 3D printer files:** https://github.com/robinhedwards/SNES-to-NeoGeo